

REQUISITIONS for lesson 1 (and /or = slides 1-40)

- Foam ball
- Tennis ball
- String or rope (for tug of war)
 - Clockwork toys/ cars

Lesson 2 / Force circus and force meters - MAKE YOUR LABELS (see Word doc) plus:

- Force meters of different sizes
- 2 magnets
- Ice cubes and wooden block
- Piece of rope
- 2 springs
- Balloon (full please)
- Plastercine
- Wooden ramp
 - toy cars
 - Tennis ball/ small ball
- Scrap paper for crumpling

Forces

Starter

1. Can we see forces?

No, but we can see the effects from them

1. What is a balanced force?

When forces are equal in size and in opposite directions

1. What is an unbalanced force?

When forces are unequal in size and cause a change in an object's speed, direction or shape

1. Draw a force diagram to show balanced forces on an object

Success criteria

Lesson Title	Working towards	Working at	Working beyond
Recap: Forces & Motion (Y7)	I can identify different forces acting on an object.	I can describe what balanced and unbalanced forces are.	I can explain how unbalanced forces change an object's motion using force diagrams.
Speed	I can identify the correct units for speed, distance, and time.	I can use the speed = distance ÷ time formula to calculate speed, distance or time.	I can solve speed questions using different units and rearranged formulas.
Describing Movement	I can read values from a distance-time graph.	I can describe motion based on the shape of the graph.	I can draw and explain distance-time graphs from written data or scenarios.
Turning Forces	I can give examples of things that turn when a force is applied.	I can use the moment formula to calculate turning forces.	I can explain how balanced and unbalanced moments affect turning.
Pressure between solids	I can state that heavier or larger objects create more pressure.	I can calculate pressure using the correct formula and units.	I can compare pressure in different situations using scientific reasoning.
Pressure in liquids and gases	I can describe how pressure changes with depth in liquids and altitude in gases.	I can explain what causes pressure in liquids and gases using particles.	I can calculate liquid pressure using the h ρ g formula and explain examples.
Particles on the move	I can define diffusion.	I can describe how temperature and concentration affect diffusion.	I can explain diffusion using the particle model and apply it to real examples.

Newton's 1st law of motion

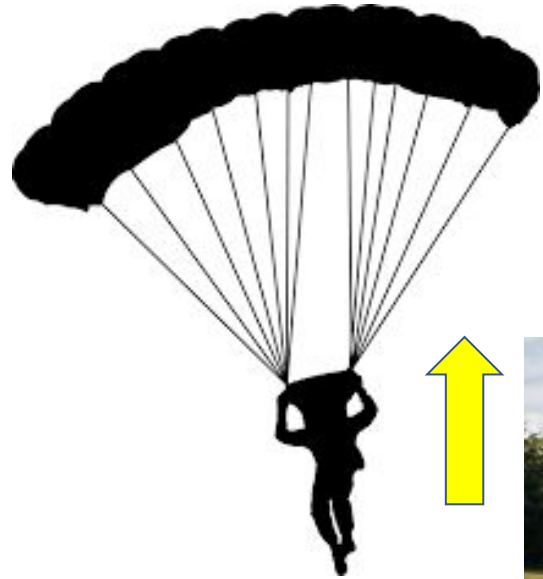


1. An object will not speed up or slow down unless an external force acts on it
2. How much an object speeds up or slows down depends upon the force applied, and the mass of the object
3. When two objects exert a force on each other, the effects are equal and opposite



We've named a few forces so far, what different types are there?





What force is this?

It is the force that resists things rubbing against each other. It acts against motion.



Friction

What force is this?

It is another force that slows things down when they are moving.



Air Resistance

What force is this?

It is the force between magnets or between a magnet and a magnetic material.



Magnetism

What force is this?

It's a force that pulls objects towards each other.

It is especially noticeable with very large objects.



Gravity

What force is this?

A strain force caused by an object being stretched.

Acts in opposite direction to the pulling force.



Tension

What force is this?

It is a force that pushes things up in water or air.



Upthrust

What force is this?

A force that slows things down when they are moving through water.



Water resistance

What force is this?

A force that is exerted by a surface on an object.



Normal contact force

What force is this?

buildup of electrical charges on the surface of materials
causes attraction/ repulsion force



Static electricity

TYPES OF FORCES

Can you categorise these forces into 2 groups?

Friction

Magnetism

Upthrust

Tension

Normal Contact Force

Gravity

Static electricity

Water resistance

Air resistance

Compare contact forces and non-contact forces. Use examples in your answer.

10:00

- Both contact forces and non-contact forces can be a push, pull or twist.
- Both contact forces and non-contact forces can change an object's shape, speed or direction.
- Non-contact forces **act at a distance** whereas contact forces need **contact with an object**.
- Examples of non-contact forces are gravity and magnetism whereas examples of contact forces are friction, air resistance, and water resistance.

To 'compare', your answer should:

- Give **similarities**.
- Write **paired statements** that **show differences relating to the same feature**.
- Use the term '**whereas**' to link your statements.

TYPES OF FORCES

Contact forces: interactions between objects that touch



applied force



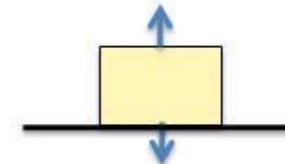
tension



drag force



frictional force

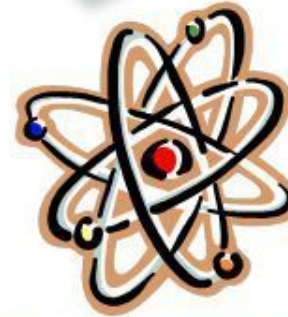


normal force

Non-contact forces: attract or repel, even from a distance



magnetic force



electric force



gravitational force

TYPES OF FORCES

Contact forces only act when two objects or materials are touching.

Examples of contact forces are:

- friction
- air resistance
- water resistance
- Upthrust
- Tension
- Normal Contact Force

Some forces can have an effect without objects touching.

They are called **non-contact forces**.

- magnetism
- gravity
- static electricity.

Investigating Forces CIRCUS EXPERIMENT

At each station read the questions and make notes in your books. In your notes include:

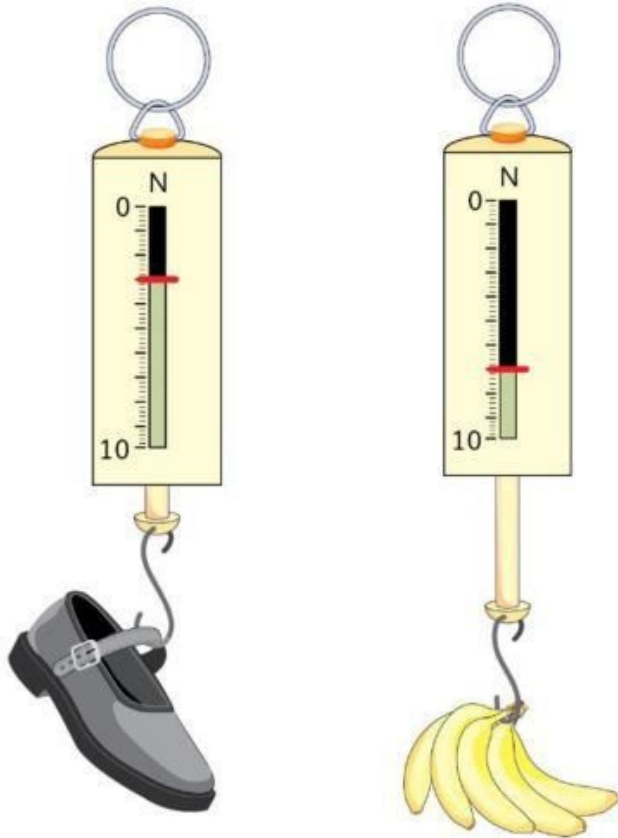
- A. Identify if forces are **balanced or unbalanced**
- B. **Draw a diagram** showing the forces (draw them large enough and neat, using a pencil)
- C. **Name/label the forces** you think are acting
- D. Identify if you saw a :
 - Change of **SHAPE**
 - Change of **SPEED**
 - Change of **DIRECTION**

Forces circus

- 1) Weigh a few objects with the force newton meter hanging from a clamp.
- 2) Crumple up one piece of paper a little the other into a ball. Drop both to the floor. What do you notice?
- 3) Stretching the spring – the force from the spring gets The more it is stretched.
- 4) Rub the balloon over your hair.
- 5) Try pulling the magnets apart
- 6) Tie knots in the string
- 7) Dragging Ice and wood
- 8) Toy car down the ramp
- 9) Make 2 different plasticine shapes and float on water

1. Weighing with a force-meter

Weigh some different objects with the force-meter
(eg. your pencil case, bottle etc.)



Force Diagram/ Free Body Diagram

2. Crumple Paper & Drop it.

Lightly crumple some paper and drop it on the floor.

Now crumple some paper really tightly and watch how long it takes to fall to the floor.

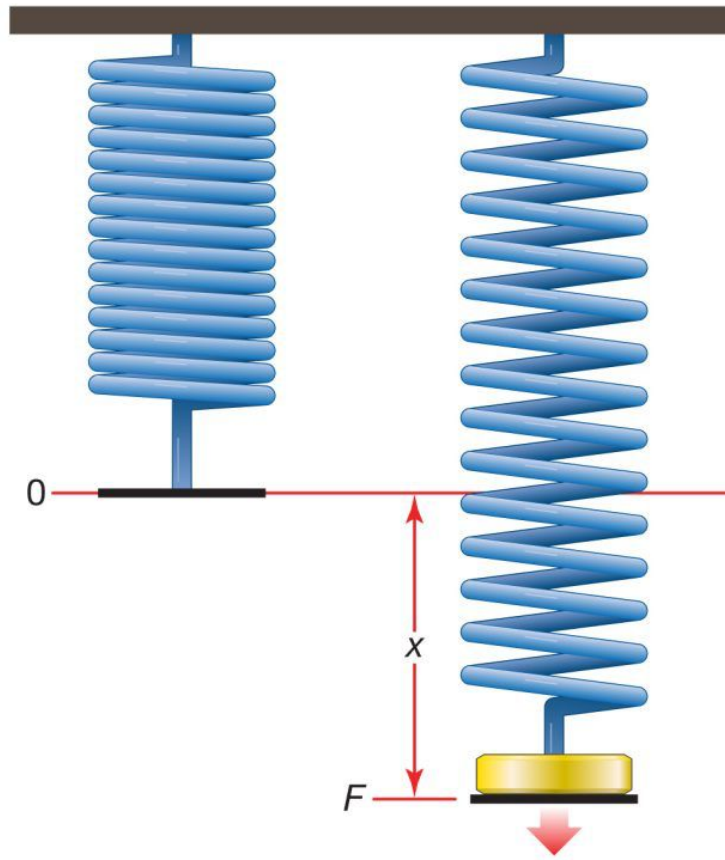
Make a note/ force diagram of which fell faster/ slower. Can you explain why?



Force Diagram/ Free Body Diagram

3. Stretching a spring

The force from the spring gets The more it is stretched.
Be Gentle! Do not overstretch the Spring!



Force Diagram/ Free Body Diagram

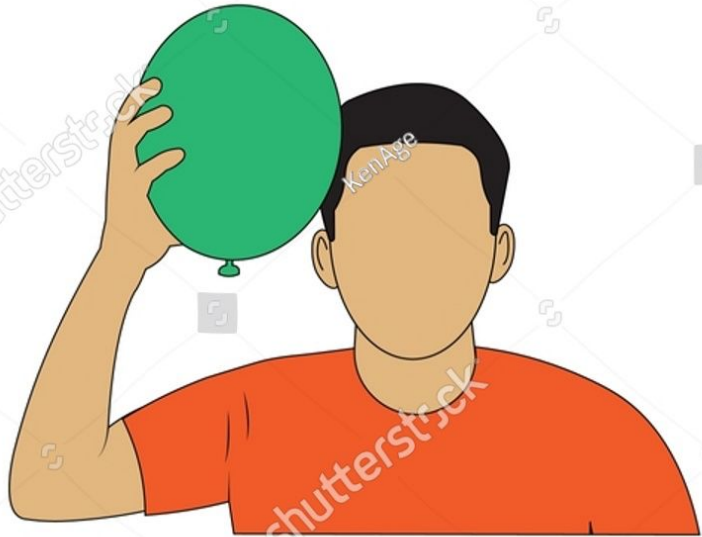
4. Rub a balloon over your hair

Step 1: Rub a balloon on your hair (if you have hair) for about 30 seconds.

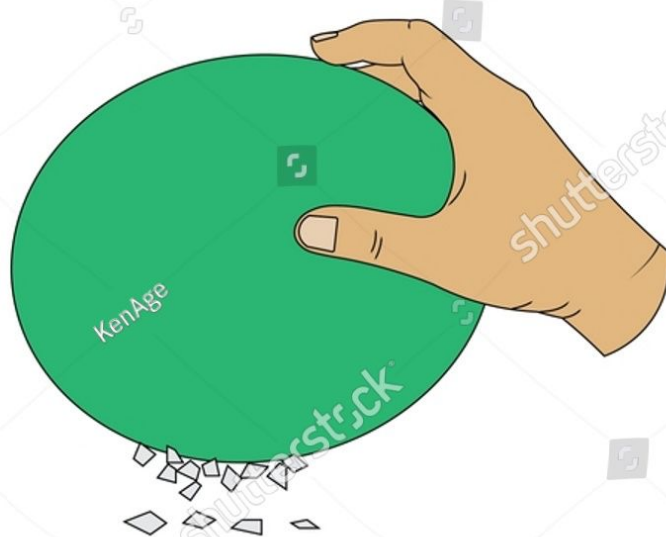
Step 2: Hover the balloon over some small pieces of paper.

Observe and name the force responsible for the phenomena shown.

Step 1



Step 2



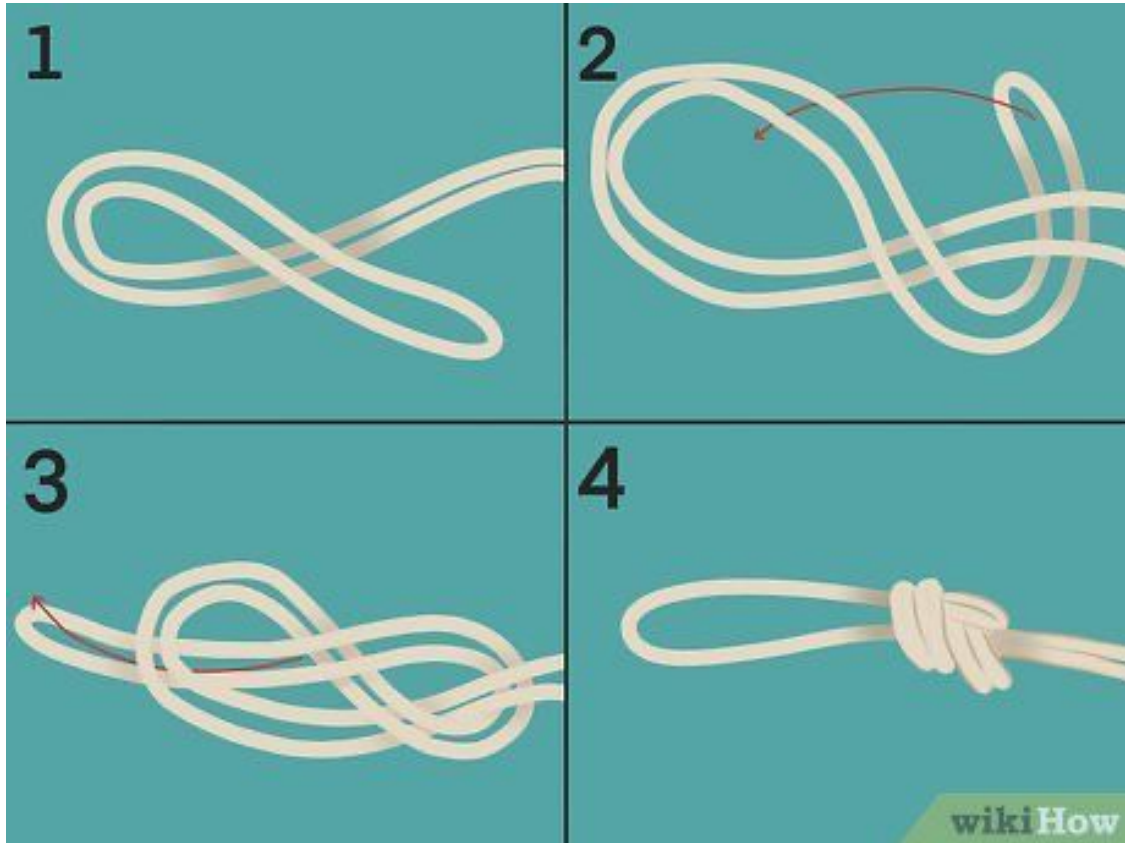
Force Diagram/ Free Body Diagram

5. Tie knots in the string

What happens when you pull it tight?

Test the knot by hanging something off of it.

Which force is responsible for keeping the know together?



Force Diagram/ Free Body Diagram

6. Magnets

Put some magnets together and pull them apart.

Now turn one of the magnets around and try to bring it toward the other magnet.



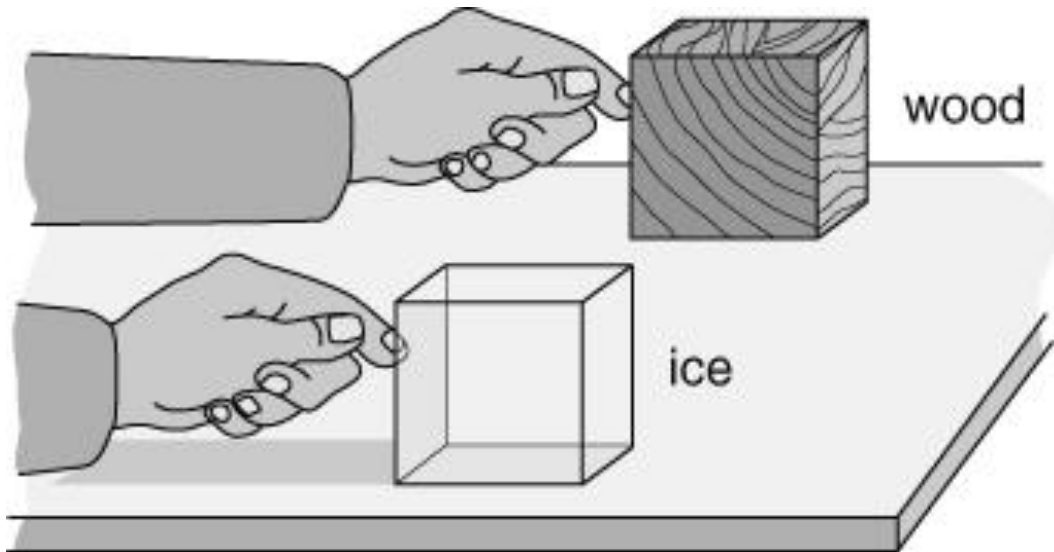
Force Diagram/ Free Body Diagram

7. Ice cube & wooden block

Push the wooden block along the bench.

Push the ice cube along the bench.

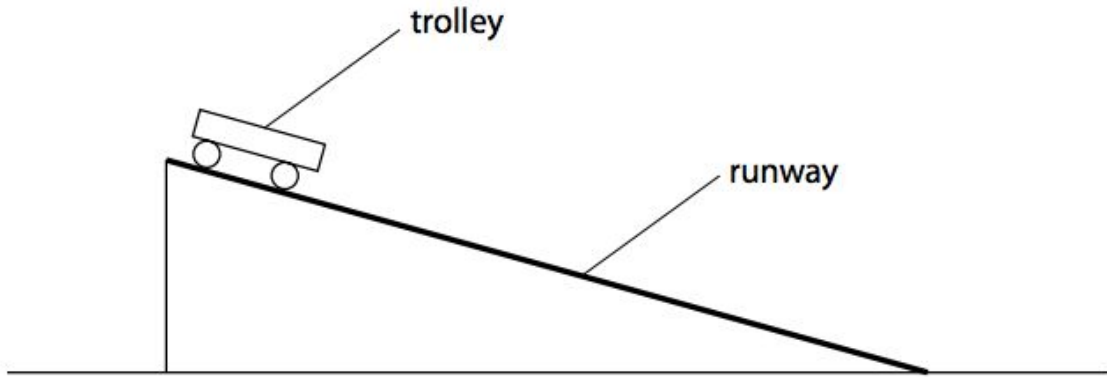
Measure the distance travelled by each of the material.



Force Diagram/ Free Body Diagram

8. Toy car down the ramp

Put the car at the top of the ramp and let it roll down



Force Diagram/ Free Body Diagram

9) Make different plasticine shapes

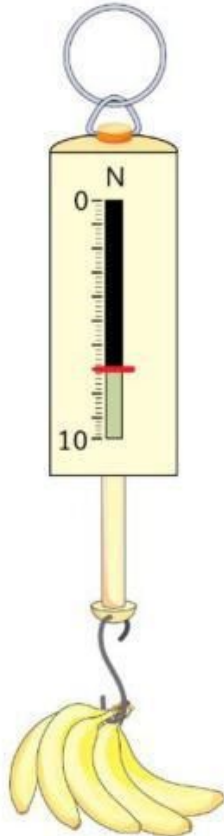
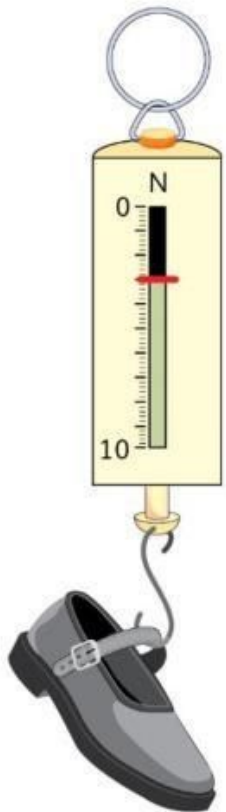


Force Diagram/ Free Body Diagram

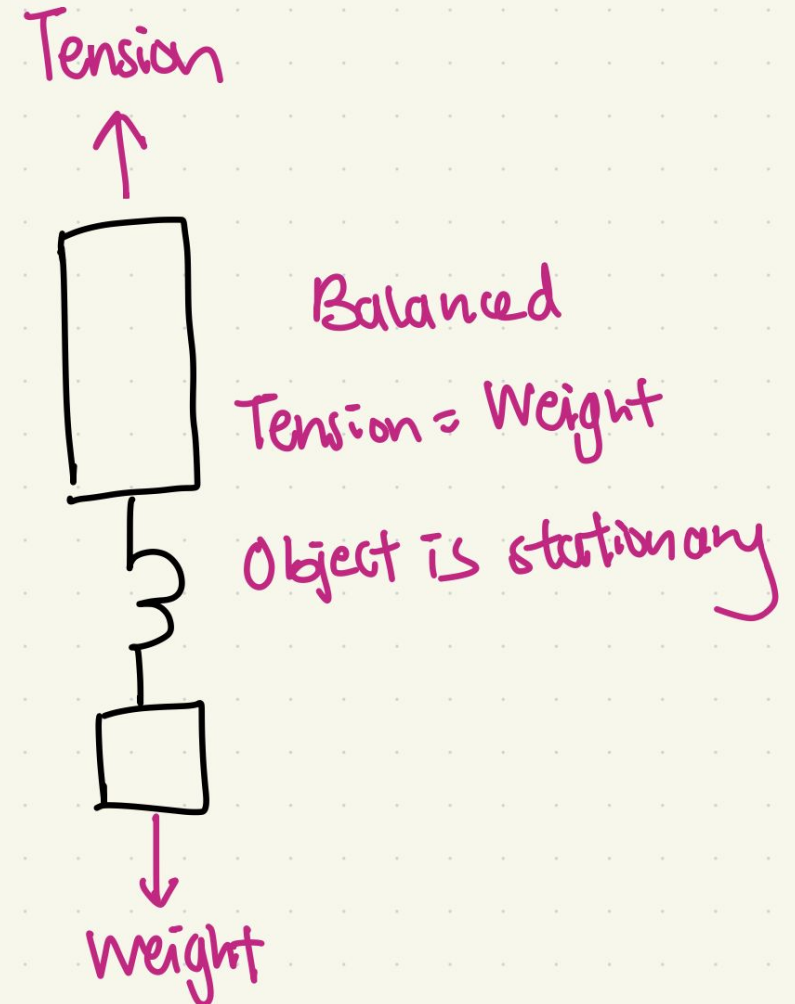
Answers from forces circus

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(eg. your pencil case, bottle etc.)



Force Diagram



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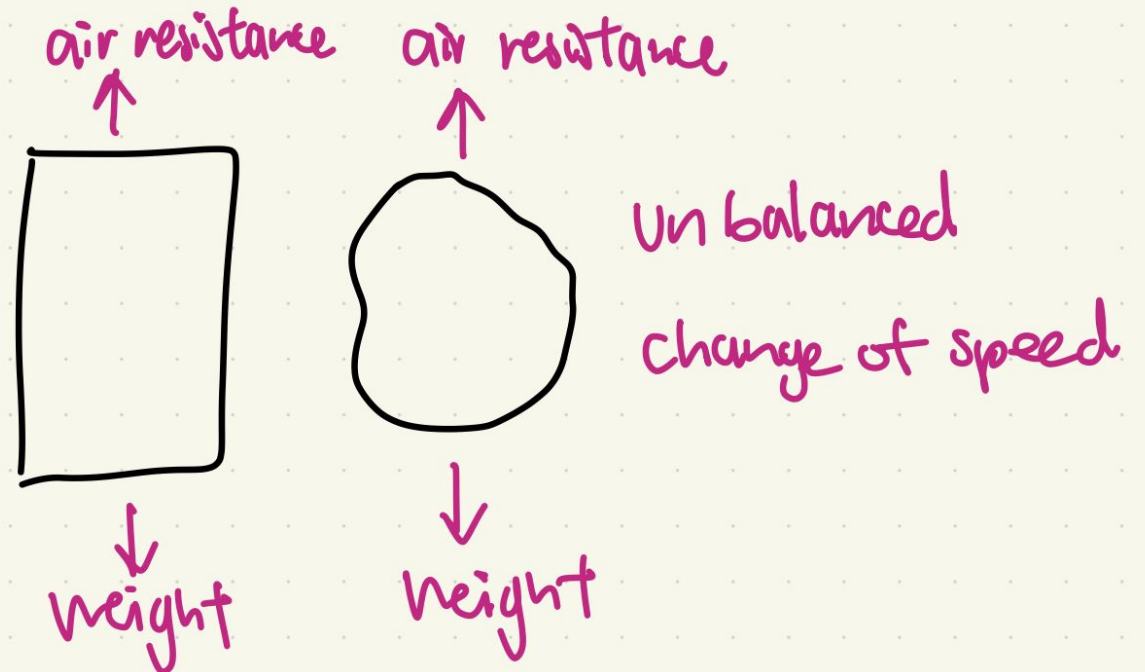
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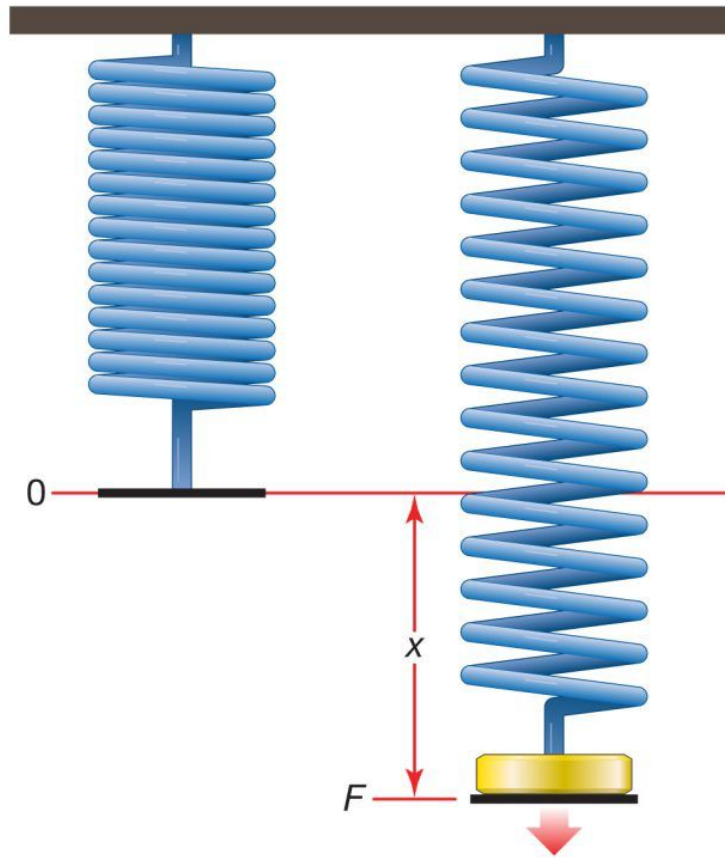


Force Diagram/ Free Body Diagram

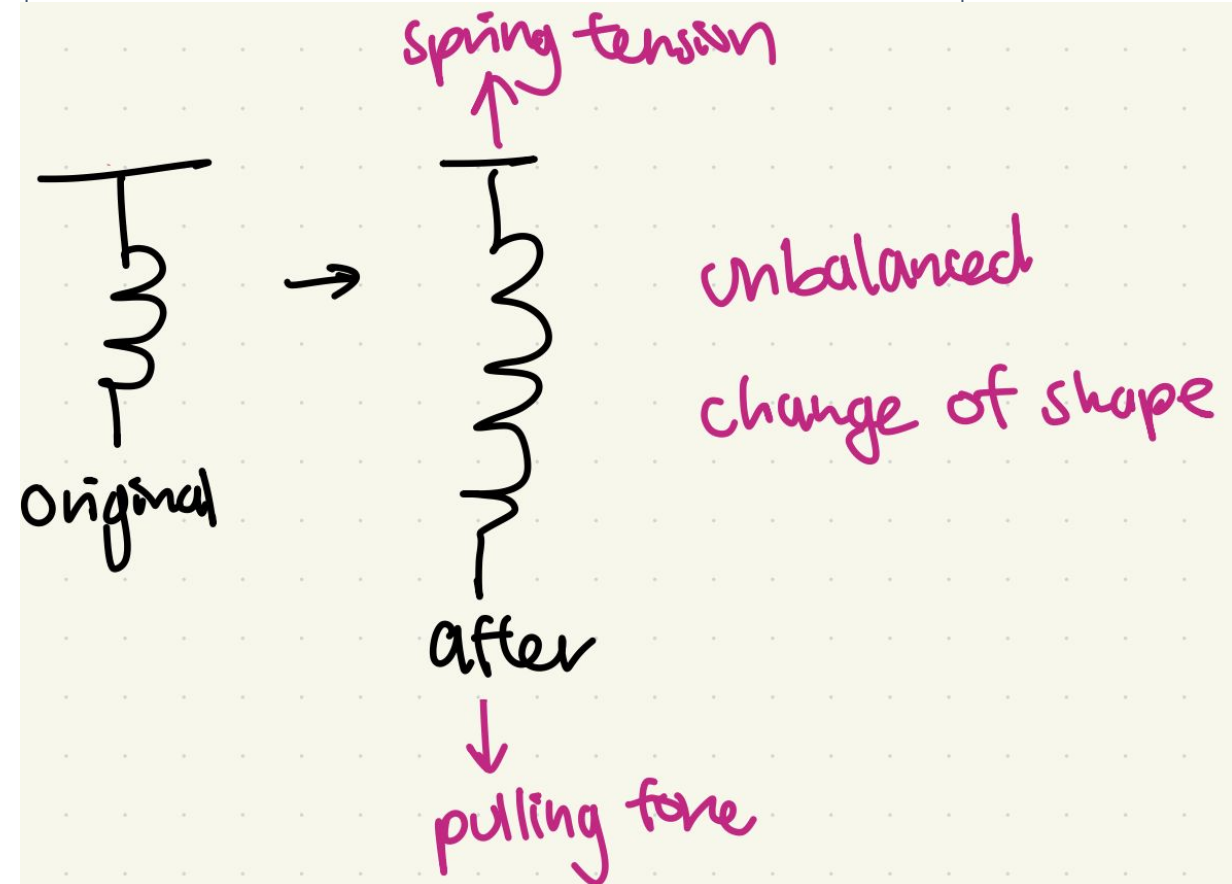


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Force Diagram/ Free Body Diagram



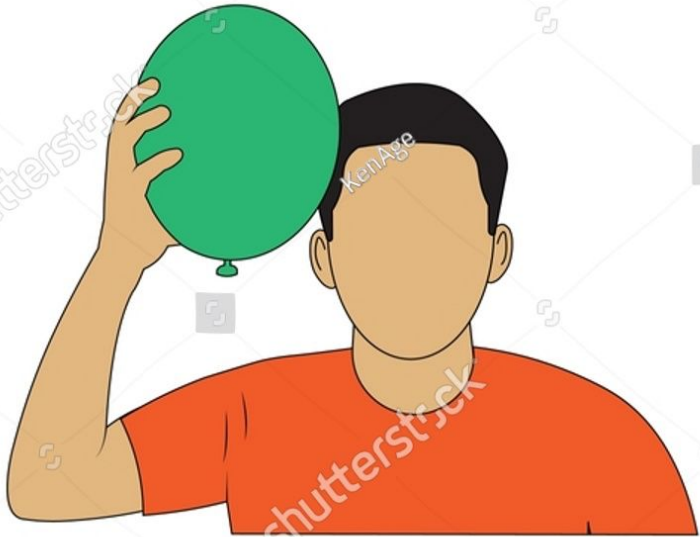
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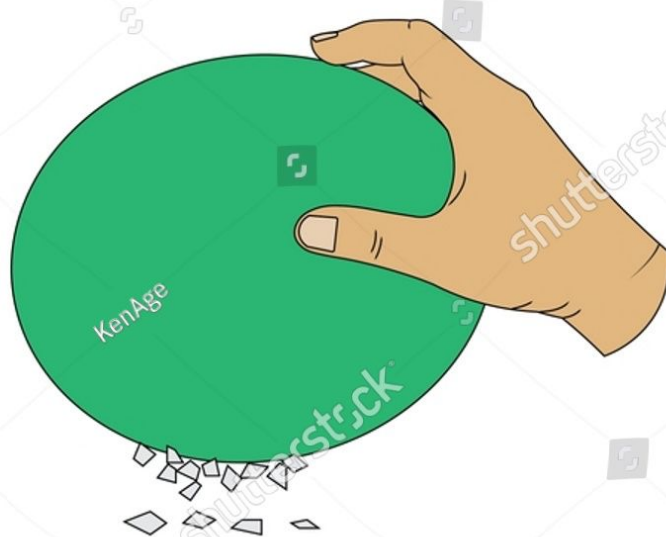
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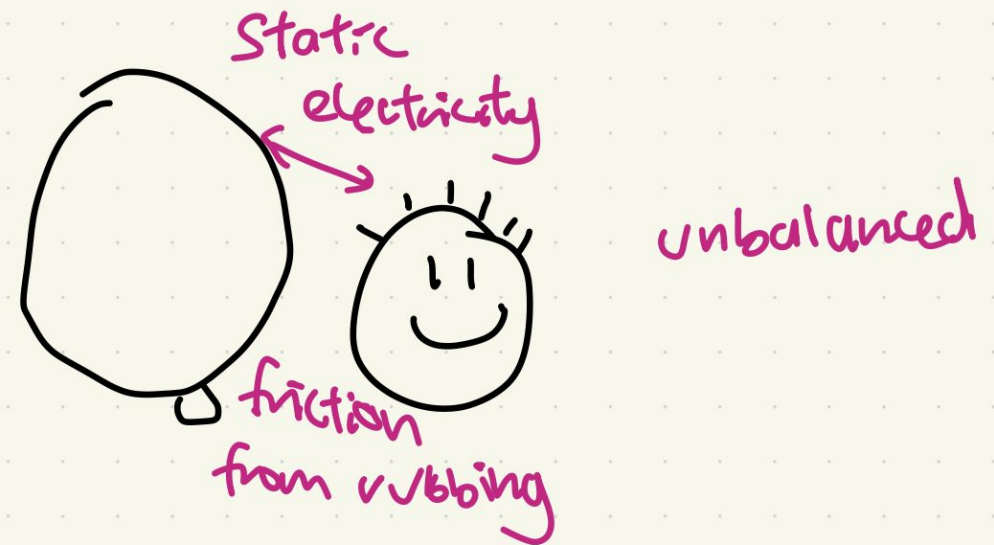
Step 1



Step 2



Force Diagram/ Free Body Diagram

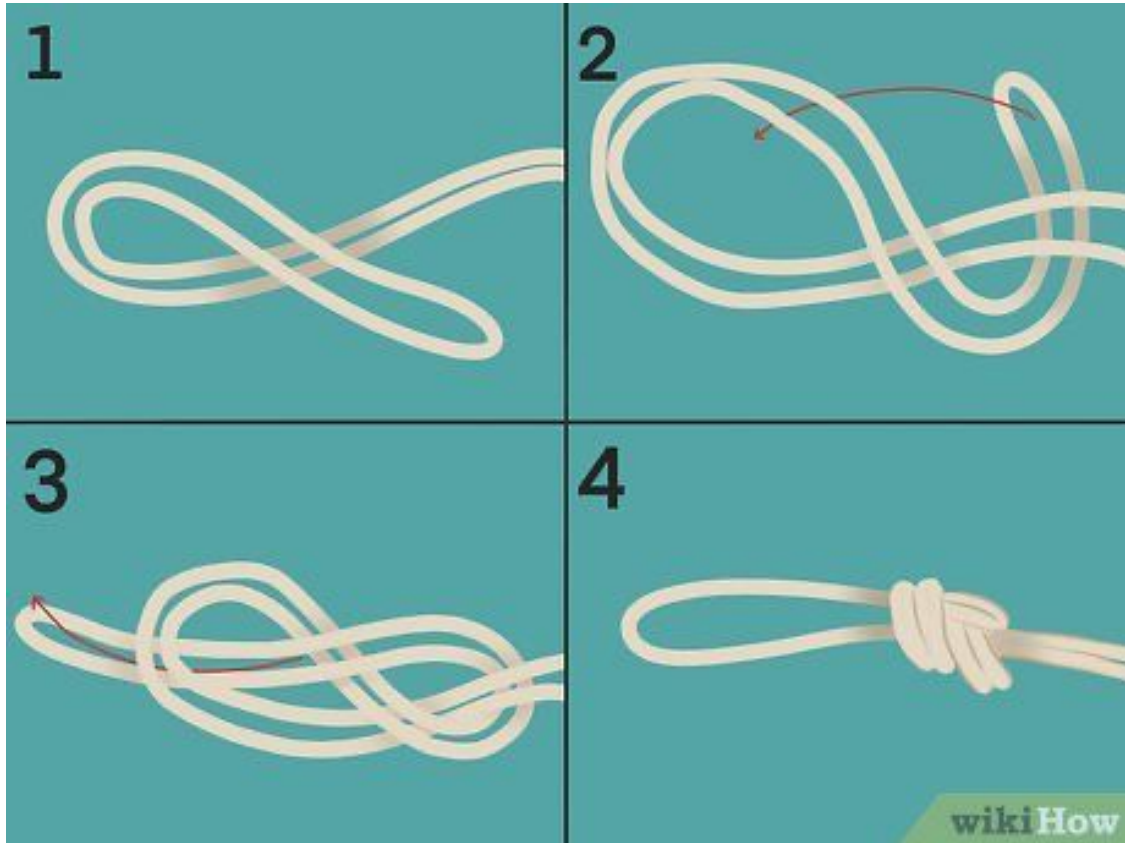


5. Tie knots in the string

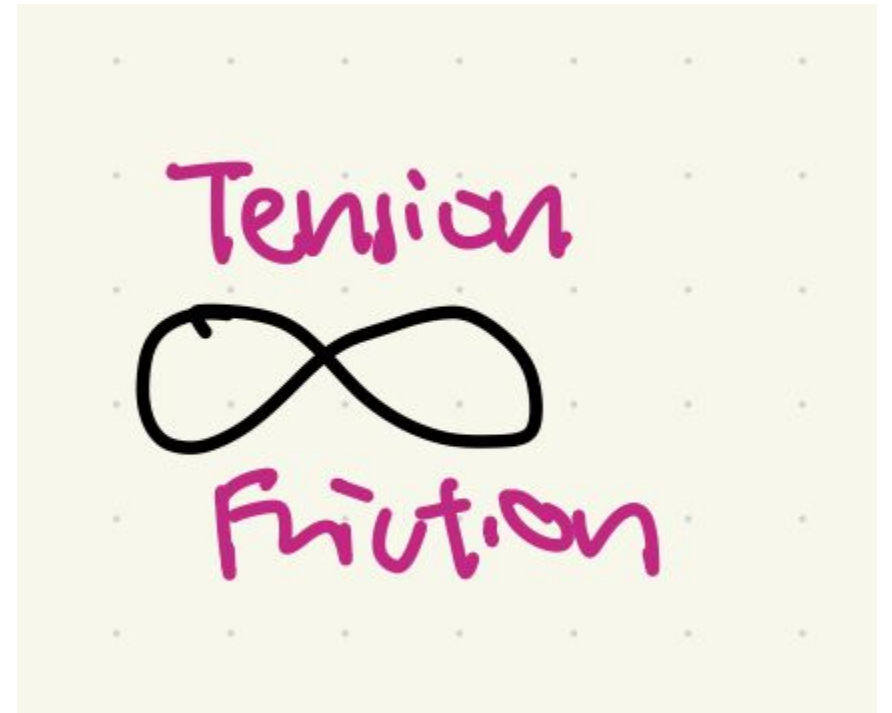
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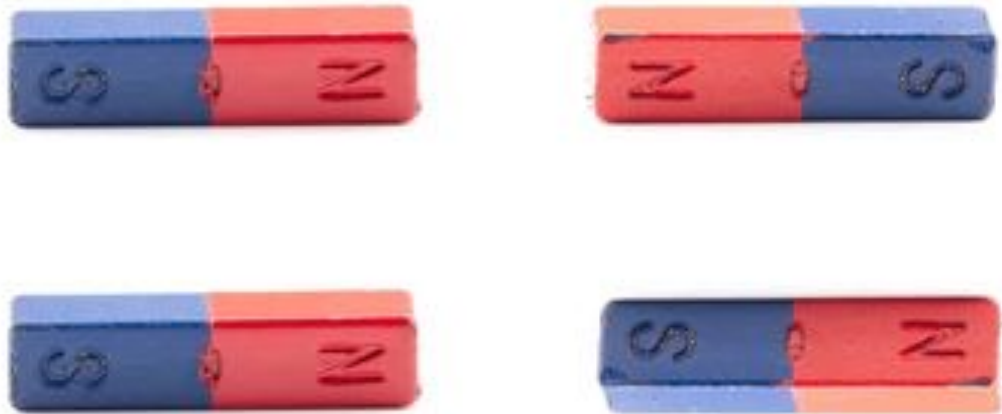
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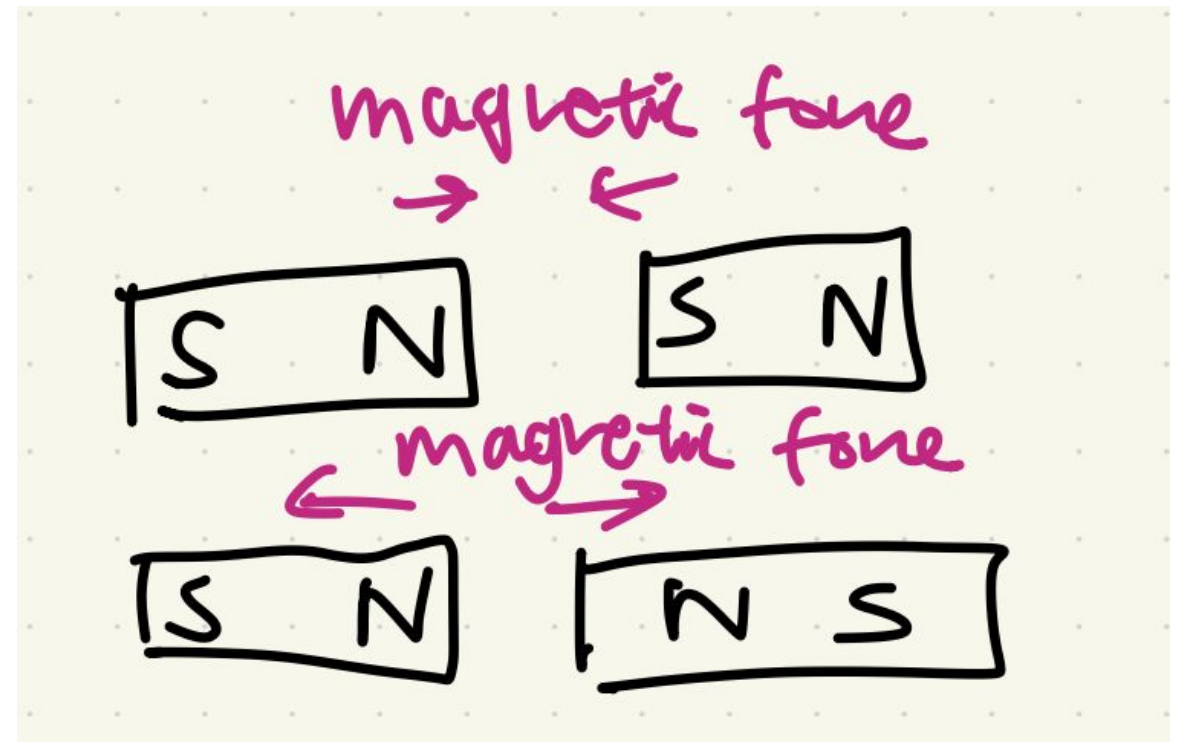
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Force Diagram/ Free Body Diagram

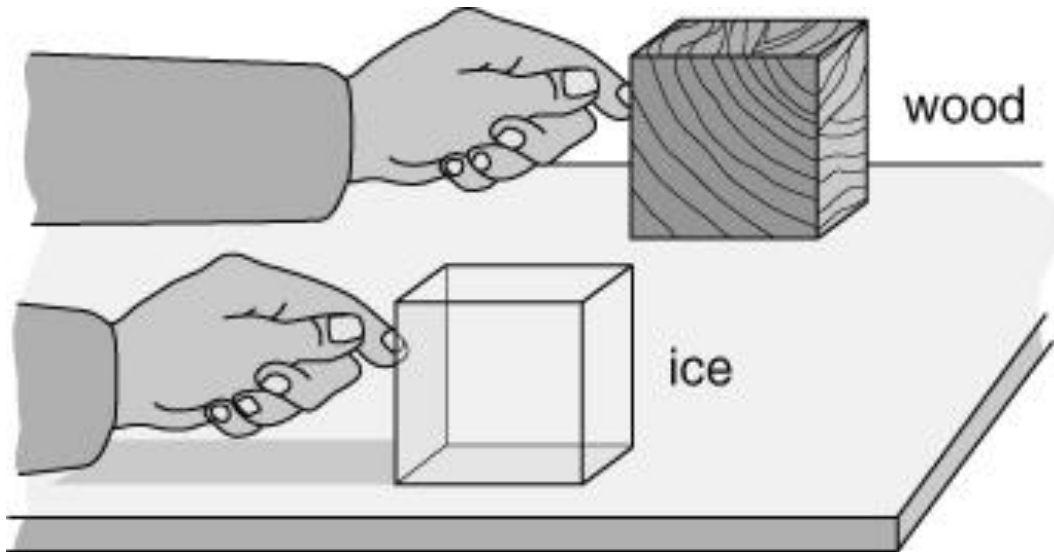


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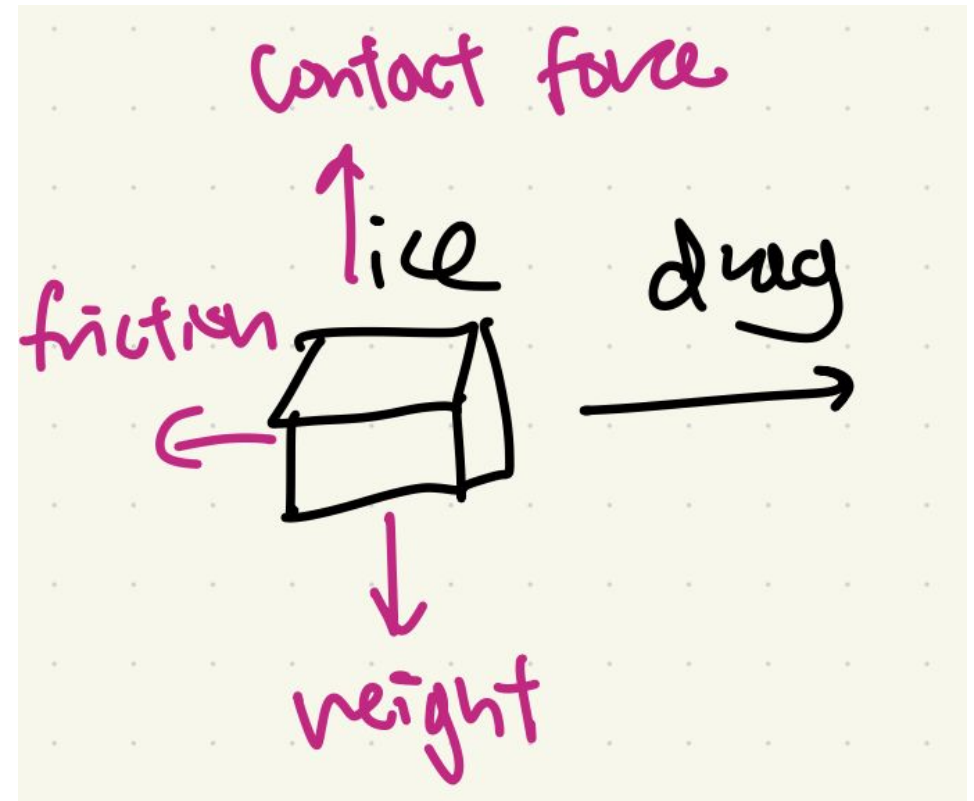
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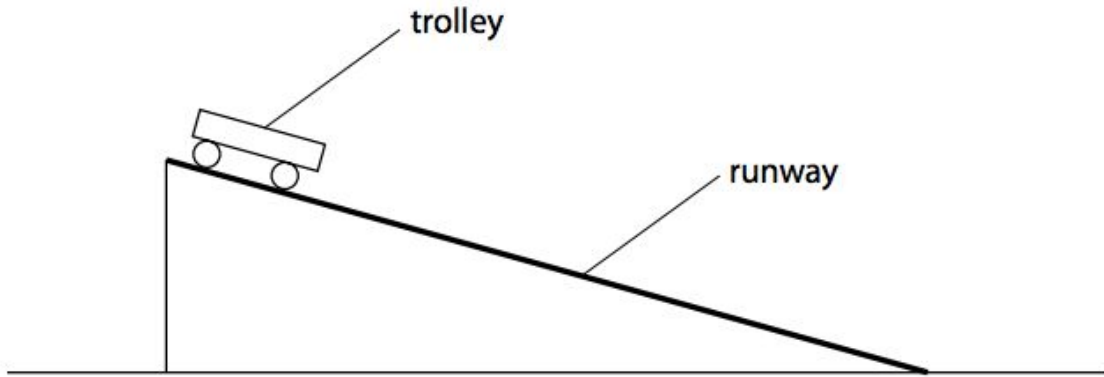


Force Diagram/ Free Body Diagram

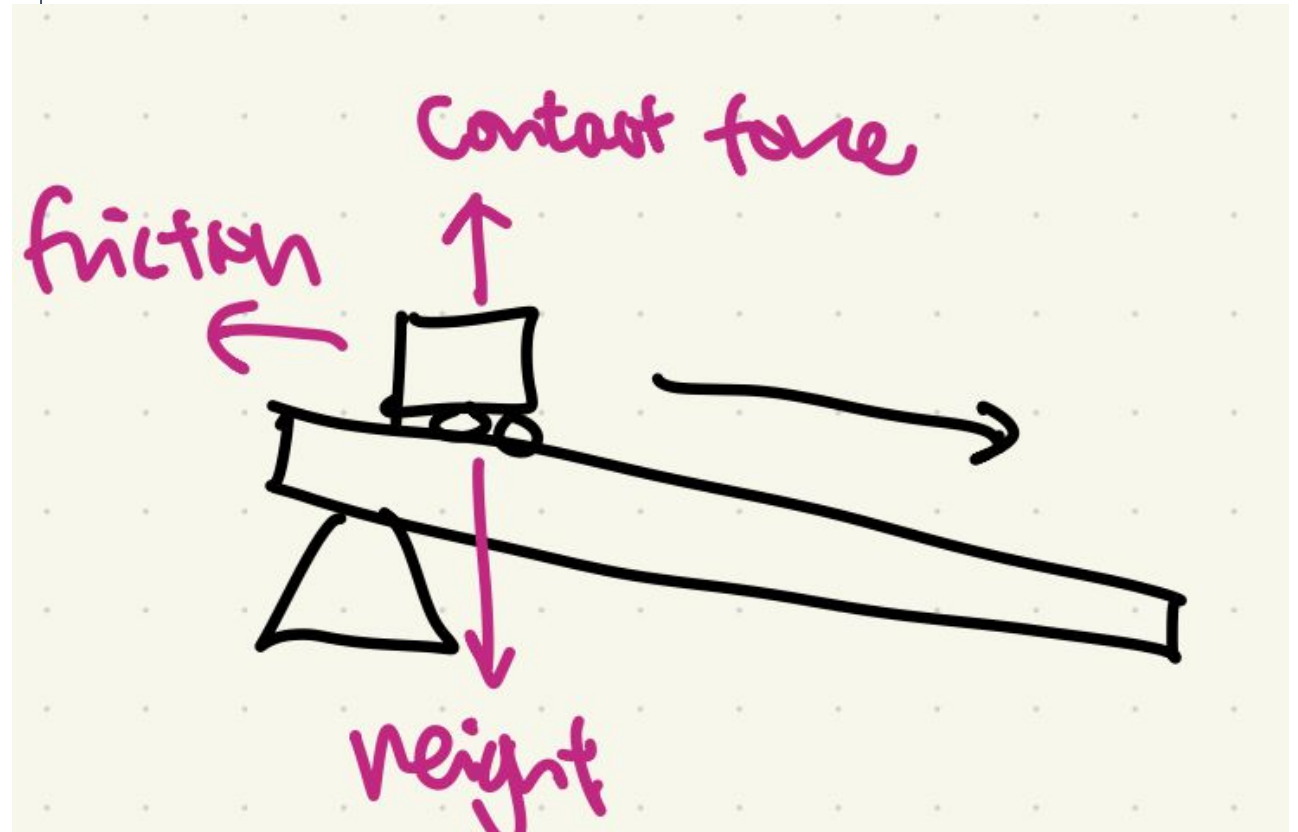


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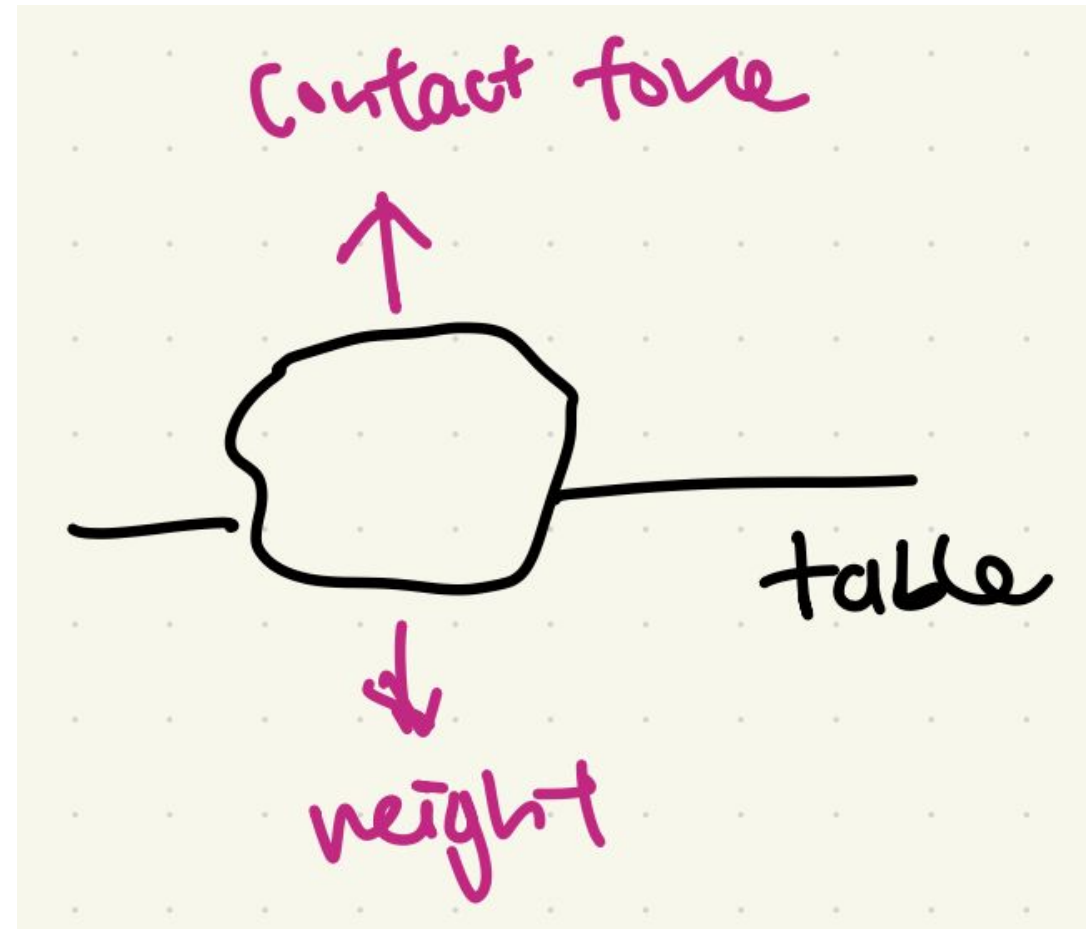
Force Diagram/ Free Body Diagram



9) Make different plasticine shapes



Force Diagram/ Free Body Diagram

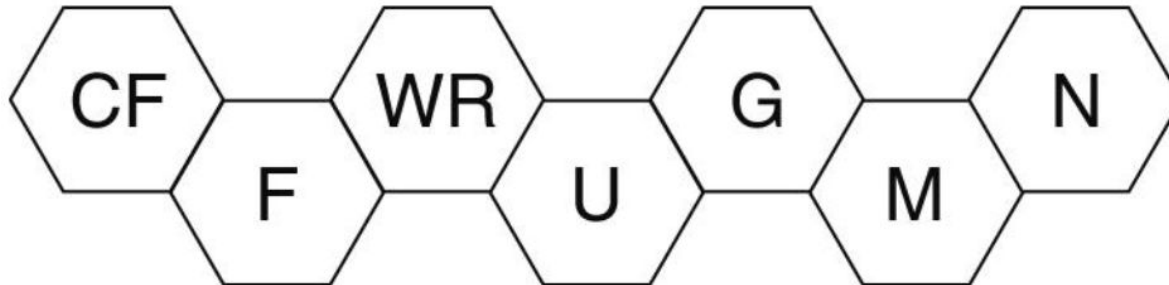




Name _____ Class _____ Date _____

1 Write down *three* things that forces can do.

2



**Complete this
worksheet**

a What CF is the name for forces where things have to be touching to have an effect?

b What F is a force that happens when two things rub against each other? _____

c What WR slows down objects moving in water? _____

d What U force makes things float? _____

Check your answers!

1 Write down *three* things that forces can do.

1. A force can change an object's direction.
2. A force can change an object's speed.
3. A force can change an objects shape and size.

Check your answers!

a What CF is the name for forces where things have to be touching to have an effect?

Contact force

b What F is a force that happens when two things rub against each other? Friction

c What WR slows down objects moving in water? Water resistance

d What U force makes things float? Upthrust

e What G is a force that pulls things downwards? Gravitational force (Gravity)

f What M is a force that attracts iron? Magnetism

g What N is the unit for force? Newton

h What M is measured in kilograms? Mass

Check your answers!

3 Write down the names of *three* contact forces.

_____ Friction, air resistance, water resistance,
_____ Upthrust, tension, Normal Contact Force

4 Write down the names of *three* non-contact forces.

_____ Magnetism, gravitational force (gravity), static
_____ electricity,
